

Differences among HTTP/1.0, 1.1, 2.0, 3.0

Respondent: 265149-周明宇

Course: Computer Networking

2026/11/03

Contents

01 HTTP basics

02 Overview of Versions

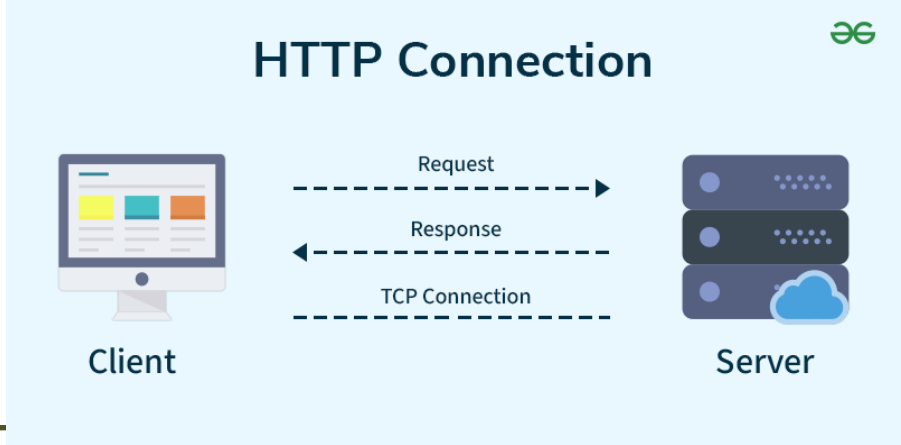
03 HTTP/1.0 vs 1.1

04 HTTP/2

05 HTTP/3 & QUIC

06 Comparison

What is HTTP?



HyperText Transfer Protocol

HTTP is the Web's application-layer protocol. A client sends a request (method, URL, headers) and a server returns a response (status code, headers, body). HTTP is stateless; cookies add state on top. Methods such as GET and status codes such as 404 have barely changed since the 1990s. Each new version changes only how these messages are carried over the network.

Overview of Versions

Version	Year / RFC	Transport & format	Key idea
HTTP/1.0	1996, RFC 1945	TCP, plain text	One request per connection
HTTP/1.1	1997, RFC 2068 (2616)	TCP, plain text	Persistent connections
HTTP/2	2015, RFC 7540	TCP, binary frames	Multiplexed streams
HTTP/3	2022, RFC 9114	QUIC over UDP	No TCP head-of-line blocking

- **Same semantics:** every version uses the same methods, status codes and headers, so websites did not need to be rewritten to upgrade.
- **What changes:** the connection model, the message format and finally the transport protocol itself.

HTTP/1.0 vs HTTP/1.1

Feature	HTTP/1.0	HTTP/1.1
Connection	New TCP connection per object	Persistent (keep-alive) by default
Requests	One at a time	Pipelining allowed, responses in order
Host header	Not required: one site per IP	Required: many sites per IP
Caching	Expires, If-Modified-Since	Cache-Control, ETag
Transfer	Whole body, length known	Chunked transfer, range requests

- **Effect:** HTTP/1.0 pays about 2 RTTs for every object; HTTP/1.1 pays the TCP handshake once and then reuses the connection, so pages with many images load much faster.

HTTP/2: same semantics, new delivery

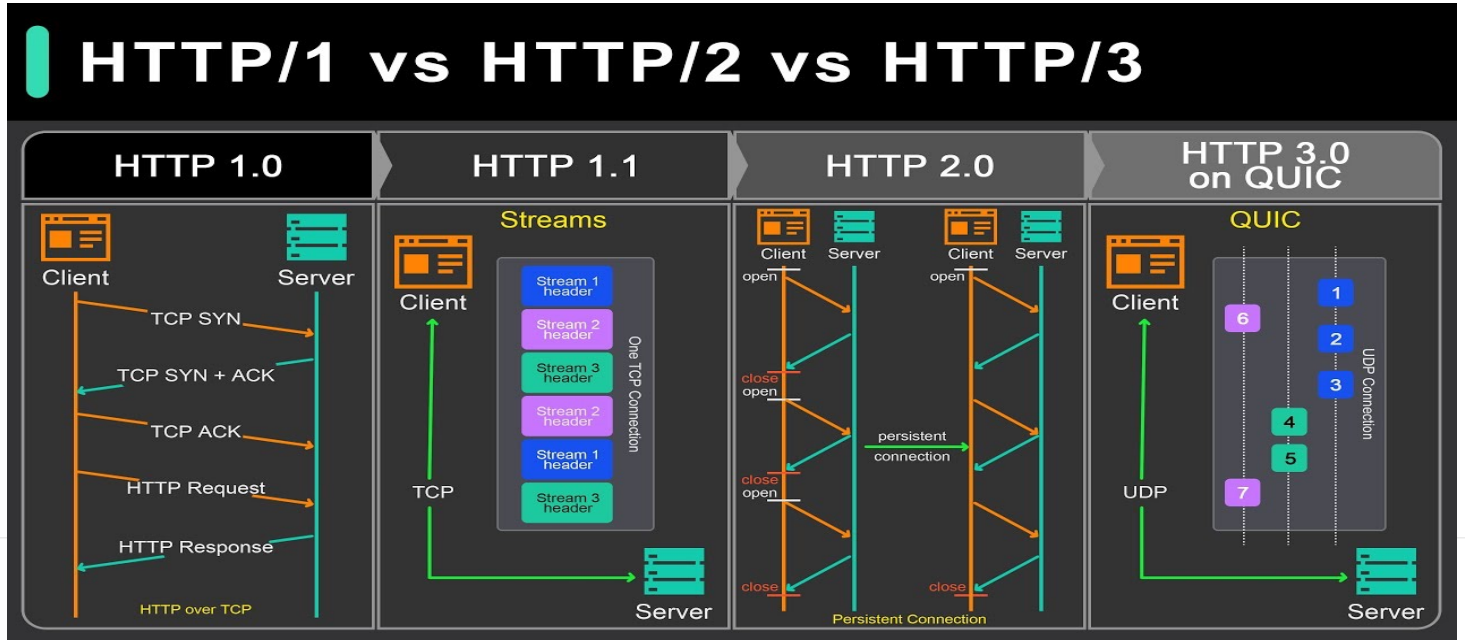
- **Binary framing:** messages are split into small binary frames that belong to numbered streams; they are faster and less error-prone to parse than text.
- **Multiplexing:** many requests and responses interleave on one TCP connection, so a large file no longer blocks small ones at the HTTP level.
- **Header compression (HPACK):** repeated headers such as cookies and user-agent are sent once and then referenced, saving bandwidth.
- **Priority and server push:** clients can mark important resources, and servers may send files before they are requested; push saw little use and browsers have dropped it.
- **Limit:** all streams still share one TCP connection, so one lost packet stalls every stream.

HTTP/3: replacing TCP with QUIC

- **New transport:** HTTP/3 runs on QUIC, a transport protocol built on top of UDP instead of TCP.
- **Independent streams:** QUIC recovers lost packets per stream, so a loss only delays the stream it belongs to.
- **Faster setup:** TLS 1.3 is built into QUIC, so a new connection needs 1 RTT and a resumed one can send data at once (0-RTT).
- **Connection migration:** a connection ID, not the IP address, identifies the connection, so it survives a switch from Wi-Fi to 4G.
- **Always encrypted:** almost all QUIC headers are encrypted, which protects privacy and stops middleboxes from interfering.

Comparison of the four versions

Version	Connection use	Head-of-line blocking	Setup RTTs	Encryption
HTTP/1.0	One per object	Each object waits for setup	1 per object	Optional
HTTP/1.1	Persistent	Responses must stay in order	1-3, once	Optional
HTTP/2	One, multiplexed	Only at TCP level	2-3, once	TLS in practice
HTTP/3	One QUIC link	Removed	1, or 0	Always



Which version is used today?

- **Wide support:** all major browsers, CDNs and large sites support HTTP/2 and HTTP/3.
- **Upgrade path:** a browser often starts on HTTP/2, learns from the Alt-Svc header or a DNS HTTPS record that HTTP/3 is available, then switches to QUIC.
- **Fallback:** if a firewall blocks UDP, the browser simply falls back to HTTP/2 over TCP.
- **HTTP/1.1 is still common:** simple APIs, internal services and tools such as curl rely on it.
- **My view:** HTTP's evolution is about latency, not new features; each version removes one more source of waiting while staying compatible.

Summary

- **HTTP/1.0:** one request per TCP connection, simple but slow.
- **HTTP/1.1:** persistent connections, Host header and better caching; still the baseline today.
- **HTTP/2:** binary frames and multiplexing on one connection, but TCP head-of-line blocking remains.
- **HTTP/3:** QUIC over UDP with built-in TLS 1.3, fixing TCP-level blocking and speeding up setup.
- **Overall:** the meaning of HTTP never changed; only the way it travels did.

Thanks for your listening!

Respondent: 265149-周明宇

Course: Computer Networking

2026/11/03